



TOR VERGATA
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Transparent by Design

A Neuro-Symbolic Approach to Mechanistic
Interpretability via Associative Memory

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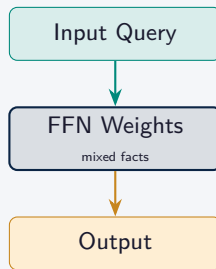
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The Core Problem: Post-Hoc Attribution Is Fundamentally Limited

Standard Transformer FFN

- Facts are **mixed** in weight matrices via gradient descent
- Attribution = inverting non-square matrix products
⇒ **ill-posed problem**
- Post-hoc tools (ROME, SHAP, Integrated Gradients) are *inherently approximate*



?
which fact?

Key Insight

The hidden nature is architectural, not just an engineering hurdle.

Our Alternative: Correlation Matrix Memories (CMMs)

A CMM stores each association as an **independent rank-1 outer product**:

$$C = \sum_{i=1}^n k_i v_i^{\top}$$

Retrieval: $\hat{v}_j = C^{\top} k_j$

Why It Matters

- Each fact is a **separate, traceable** matrix term
- Under key independence, retrieval is **exact**
- Attribution is a **closed-form algebraic** operation

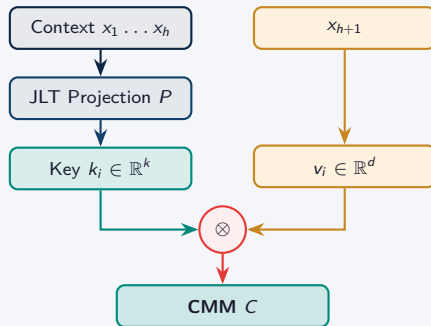
$$C = \begin{matrix} \boxed{k_1 v_1^{\top}} \\ \boxed{k_2 v_2^{\top}} \\ \boxed{k_3 v_3^{\top}} \\ \boxed{k_n v_n^{\top}} \end{matrix} \left. \vphantom{\begin{matrix} \boxed{k_1 v_1^{\top}} \\ \boxed{k_2 v_2^{\top}} \\ \boxed{k_3 v_3^{\top}} \\ \boxed{k_n v_n^{\top}} \end{matrix}} \right\} \begin{matrix} \text{Independent} \\ \text{rank-1 terms} \end{matrix}$$

The MeMo Architecture: CMMs for Language Modeling

MeMo (Zanzotto et al., 2025) extends CMMs to **sequence modeling**:

1. Tokens $\rightarrow$ Gaussian random vectors $\sim \mathcal{N}(0, I_d)$
2. Context concatenated & compressed via **JLT**:
 $P \in \mathbb{R}^{k \times (h \cdot d)} \rightarrow \text{key } k_i$
3. Next token $\rightarrow$ value v_i
4. Stored via outer product: $C \leftarrow C + k_i v_i^\top$
5. Layer ℓ memorizes sequences of length ℓ

Our contribution: We show this memory can be *checked*, *edited*, and *erased* with closed-form math.



Three Pillars of Our Framework

DIAGNOSE

Transparency Index $\mathcal{T}$

Closed-form attribution: trace
outputs to specific stored facts

INJECT

Symbolic Rule Injection

Inject $A \Rightarrow B$ via $C + \alpha \cdot k_A v_B^\top$

No gradients needed

ERASE

Exact Unlearning

Remove facts via $C - \alpha \cdot k_{\text{err}} v_{\text{err}}^\top$

Zero collateral damage*

All three: **gradient-free, deterministic, closed-form**

* Under key independence (JLT guarantee)

Pillar 1: The Transparency Index $\mathcal{T}$

Decomposition. Every output $\hat{v} = C^\top s$ decomposes as:

$$\hat{v} = \sum_{i=1}^n \underbrace{(s^\top k_i)}_{\text{similarity}} \underbrace{v_i}_{\text{stored value}} = \sum_i c_i$$

Definition:

$$\mathcal{T}(\hat{v} \mid s) = \frac{\max_i \|c_i\|}{\sum_j \|c_j\|}$$

- $\mathcal{T} = 1$: perfect attribution (single fact)
- $\mathcal{T} \approx 1/n$: uniform spread $\rightarrow$ hallucination indicator
- Bounded: $\mathcal{T} \in [1/n, 1]$

Theorem: Distinguishing Gap (JLT)

For memorized query $s = k_m$ vs. OOD query $s_\perp$:

$$\mathbb{E}[\mathcal{T}(s)] - \mathbb{E}[\mathcal{T}(s_\perp)] \geq \frac{1}{4} \cdot \frac{1 - (n-1)\epsilon^2}{n}$$

with probability $\geq 1 - O(n^{-1})$.

Memorized keys dominate the dot-product sum;
OOD keys spread weight uniformly.

Pillar 2: Symbolic Rule Injection

Encode a propositional rule $A \Rightarrow B$ directly:

$$C^{(\text{last})} \leftarrow C^{(\text{last})} + \alpha \cdot k_A v_B^\top$$

Key properties:

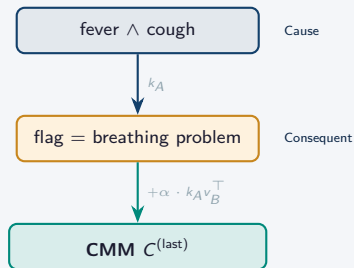
- **Gradient-free:** pure matrix addition
- **Attribution-complete:** $\mathcal{T}$ can trace the rule
- **Collateral-free** (under JLT):
small change $\|\Delta_i\| \leq \epsilon \alpha \|v_B\|$

Contrast with ROME

ROME degrades specificity to 62.1% on GPT-2 XL.

Our injection: 100% compliance, $\Delta\mathcal{T} < 10^{-4}$.

Clinical Decision Example



10/10 rules $\rightarrow$ 100% compliance
at $\cos \geq 0.85$ threshold

Pillar 3: Algebraically Exact Unlearning

Remove a stored association $(k_{\text{err}}, v_{\text{err}})$:

$$C_{\text{new}} = C_{\text{old}} - \alpha \cdot k_{\text{err}} v_{\text{err}}^{\top}, \quad \alpha = \|k_{\text{err}}\|^{-2}$$

Proposition: Exactness Condition

If $k_{\text{err}} \perp k_j$ for all remaining keys j :

- Retrieval of erased fact $\rightarrow$ **exactly zero**
- All other associations $\rightarrow$ **unchanged**

Under **approximate independence** (JLT), residual interference is bounded by ϵ .

$k_1 v_1^{\top}$ ✓

$k_2 v_2^{\top}$ ✓

$k_{\text{err}} v_{\text{err}}^{\top}$ ✗

$k_4 v_4^{\top}$ ✓

-- $\rightarrow$ Subtract
 $\alpha \cdot k_{\text{err}} v_{\text{err}}^{\top}$

Retrieval $\rightarrow < 10^{-5}$
Collateral $\Delta \mathcal{T} < 10^{-4}$

Scaling: Hierarchical Tree Routing

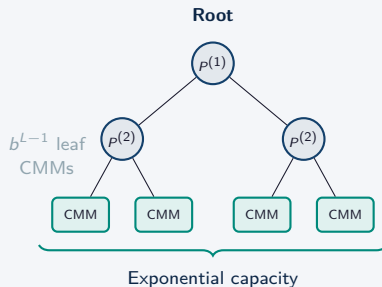
Single-layer capacity: $n \approx \beta \cdot N_{\text{oP}}$

At $d=2048$: $\sim 250\text{K}$ sequences with 3 layers

Hierarchical scaling: tree of CMMs with branching factor b :

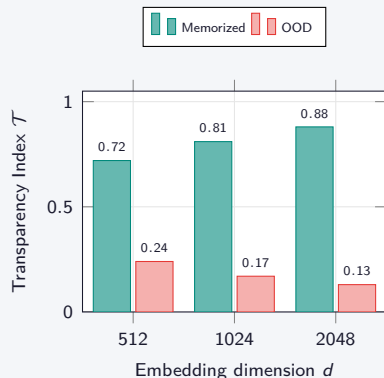
$$N_{\text{total}} = b^{L-1} \cdot \beta \cdot N_{\text{oP}}^{(\ell)}$$

- Routing margin: $\gamma > \Omega(\sqrt{(\log b)/d})$
- **No compounding crosstalk:** each leaf has local keys
- $\mathcal{T}$ also serves as *capacity health indicator*



Pilot Experiments: Key Results

LAMA Fact Retrieval (200 facts, T-REx)



Summary of Pilot Results

Metric	Result
Fact retrieval	100% Precision@1
$\mathcal{T}$ (memorized, $d=2048$)	0.88 ± 0.02
$\mathcal{T}$ (OOD, $d=2048$)	0.13 ± 0.03
Rule compliance	10/10 (100%)
Post-unlearn retrieval	$< 10^{-5}$
Collateral damage	$\Delta\mathcal{T} < 10^{-4}$

✓ Results consistent with Theorem 1: gap widens with d .

How We Compare: CMM vs. Model-Editing Baselines

Property	Ours (CMM)	ROME	MEND	GRACE
Gradient-free	✓	×	×	×
Closed-form edit	✓	×	×	×
Attribution-complete	✓	×	×	×
Exact unlearning	✓*	×	×	×
No collateral damage	✓*	×	×	✓
Scales to LLM size	Open	✓	✓	✓
Paraphrase-robust	Open	✓	✓	✓

*Under the key independence condition guaranteed by the JLT.

Limitations & PhD Roadmap

Current Limitations

- Single-layer CMM (not full multi-layer MeMo)
- Small benchmark: 200 LAMA facts + 10 rules
- Hierarchical routing not yet validated practically
- Exact grammatical match only, no paraphrase robustness yet

Next Steps (PhD Roadmap)

- Scale to full **CounterFact** (21,919 edits)
- Benchmark vs. ROME, MEND, GRACE on GPT-2 Medium
- Validate hierarchical routing error rates
- Build **PyTorch** / **HuggingFace** wrappers
- Extend to **semantic** / **paraphrase** key matching



Key Takeaway

Don't explain a black box after the fact.
build a **transparent** one from the start.

Three closed-form operations on CMM-based memory:

$\mathcal{T}$ -Index for attribution ■ + for injection ■ - for unlearning

Thank You & Questions

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